CPU Scheduling Simulator

Operating Systems Lab Assignment

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1. Introduction

CPU Scheduling is an important concept in operating systems. It determines how the CPU selects a process from the ready queue for execution. Efficient scheduling improves system performance, reduces waiting time, and increases CPU utilization.

This project implements a CPU Scheduling Simulator that demonstrates different scheduling algorithms and compares their performance.

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2. Objectives

- To understand CPU scheduling algorithms
- To implement different scheduling techniques
- To calculate Waiting Time and Turnaround Time
- To visualize execution using Gantt Chart
- To compare performance of different algorithms

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3. Algorithms Implemented

3.1 First Come First Serve (FCFS)

- Processes are executed in the order of arrival

- Simple and easy to implement
- May cause long waiting time (convoy effect)

3.2 Shortest Job First (SJF) – Non-Preemptive

- Process with smallest burst time is executed first
- Reduces average waiting time
- Can cause starvation for longer processes

3.3 Round Robin (RR)

- Each process is given a fixed time quantum
- Ensures fairness among processes
- Suitable for time-sharing systems

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4. Performance Metrics

Turnaround Time (TAT)

Turnaround Time = Completion Time – Arrival Time

Waiting Time (WT)

Waiting Time = Turnaround Time – Burst Time

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5. Output Results

(Screenshots are attached in the screenshots folder)

- FCFS Output
- SJF Output
- Round Robin Output

Each output shows:

- Completion Time
- Turnaround Time
- Waiting Time
- Gantt Chart

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6. Challenging Task

6.1 Shortest Remaining Time First (SRTF)

- Pre-emptive version of SJF
- Process with smallest remaining burst time is selected
- If a shorter process arrives, current process is interrupted

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6.2 Comparison of Algorithms

Algorithm		Avg Waiting Time	Avg Turnaround Time
FCFS		2.0	6.5
SJF		4	9.25
RR		8.5	13.75
SRTF		7.3	12.33

7. Conclusion

This project helped in understanding how different CPU scheduling algorithms work.

- FCFS is simple but not efficient
- SJF provides better performance but may cause starvation

- Round Robin ensures fairness
- SRTF gives optimal waiting time but is complex

Overall, SJF and SRTF provide better performance in terms of waiting and turnaround time.

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8. How to Run the Program

1. Open terminal
2. Run the command:
   python main.py
3. Enter process details as required
4. Select the scheduling algorithm
5. View output and Gantt Chart

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9. Tools Used

- Python
- VS Code